

Auctions with Risk Reserves for Scarce AI Research

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KEY FINDING

Reserving one of four AI research slots for a verified high-risk request increases risk coverage but displaces private research value. In our baseline simulation, the hybrid improves mean welfare only when the firm's risk weight exceeds the break-even level $\lambda^* \approx 0.733$.

01 Question & model

Research question & verified gap

Can one verified high-risk reserve improve scarce AI research allocation relative to a pure first-price auction(FPA)?

Gap: Prior work studies GenAI in asset management, compute-resource markets, and non-price reserves separately[1–3].

Strategic model & foundations

- 20 PMs compete for 4 slots; at most one per PM. Private value $v \in [0,100]$; budget = 100 credits.
- Public risk score: $r_i = 100E_iS_i$, where exposure E_i and severity $S_i \in [0,1]$.
- Risk scores are frozen before bidding and cannot be manipulated within the batch.
- Requests with $r_i \geq 60$ are reserve-eligible.
- Unspent credits retain outside value; this is one static batch.

Three complementary lenses

- Game theory:** Private values and incomplete information generate strategic bid shading.
- Mechanism design:** a free reserve changes access and displacement.
- Social choice:** $W = V + \lambda C$; V = total private research value C = total verified risk coverage

when does the reserve help?

$$\Delta C = C(H) - C(P); L = V(P) - V(H)$$

$$W(H) > W(P) \Leftrightarrow \lambda \Delta C > L$$

Payments are internal transfers; PM utility is $u = xv - p$.

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Professor Luyao Zhang: poster template.

OpenAI ChatGPT / Codex assisted with model, code, literature and writing; authors retain responsibility.

Python: python3 simulation.py

6 tests passed; 31,265 output cells verified.

02 Design & evidence

MATCHED REQUESTS, THREE RULES

20 PMs → same values, risk scores and bids

PURE FPA

4 highest bids win;
each pays own bid.

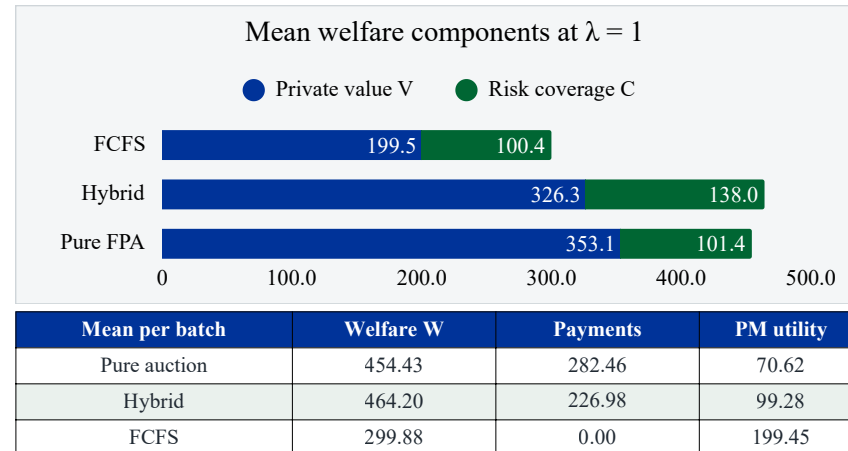
HYBRID

Highest $r \geq 60$: free slot.
Then 3 auction slots.

FCFS

4 earliest requests;
exogenous order, free.

1,000 independent batches; seed 20260926. Fixed $b = 0.8v$ in both auctions (an assumed rule, not an equilibrium). If no $r \geq 60$, auction all 4 slots.



HYBRID VS. PURE FPA

$\Delta C = +36.59$ risk coverage; $L = 26.82$ displaced private value

Mean $\Delta W = +9.78$ at $\lambda = 1$; MCSE = 0.89

Only 43.8% of batches have $\Delta W > 0$: The hybrid improves mean welfare, not every allocation.

SELECTED REFERENCES

- [1] Sheng et al. (2026). Generative AI and Asset Management. Review of Financial Studies. doi:10.1093/rfs/hhag050
- [2] Sivan et al. (2026). Quota Marketplace: Dynamic Pricing for Efficient Allocation of ML Training Resources. arXiv:2607.09802.
- [3] Pathak et al. (2023). Fair Allocation of Vaccines, Ventilators and Antiviral Treatments: Leaving No Ethical Value Behind in Health Care Rationing. arXiv:2008.00374.

03 Impact & dialogue

Practical impact: a real-world case

- T. Rowe Price's Claude rollout to PMs and analysts motivates the setting.
- The 20-PM / 4-slot constraint is our research assumption, not a reported firm policy.
- Risk coverage may support accountable allocation (SDG 16.6); it is not measured avoided loss.

Interdisciplinary Evidence

- Behavioral test:** Four bidders compete for one slot under equivalent gain/loss wording.
- Rational single-slot benchmark:** $b(v) = 0.75v$
- Status:** No human rounds have yet been completed; peer-play evidence remains planned.
- Decision rule:** If observed bids systematically depart from $0.75v$, we test whether importing the observed bid ratio changes the ranking of Pure FPA and Hybrid.

Limitations & next test

- Mean outcomes are conditional on the welfare weight λ and the assumed bidding rule.
- Risk coverage is a policy proxy, not measured avoided financial loss.
- Next: behavioral evidence, formal equilibrium analysis and a shadow-allocation audit.

Takeway

- Pure FPA prioritizes private research value.
- The risk reserve prioritizes verified downside exposure.
- The hybrid is justified only when the added risk coverage is worth the displaced private value.

OPEN MATERIALS

[GitHub: code + notebook](#)

[JianNi-220/PS2_Qianshuo_Zhenning](#)

[Hugging Face: live game](#)

[Zn7777/PS2_Qianshuo_Zhenning_PM_Bid](#)